

Objectives

Recognize four emotion states: Valence, Arousal, Dominance, Liking through three neural network models on DEAP dataset
Motivations: 1. Clinical psychiatric disorder diagnosis, 2. Improving emotional features in Human-Computer Interaction

Background:

- EEG recordings are widely used in clinical practice.
- Comprise the signals originating from the central nervous system (CNS)
- Include emotional information

Dataset overview:

- 32 subjects (16 male, 16 female, 19–37 years old)
- 40 trials of one-minute music videos
- (3s baseline + 60s stimulus for each trial)

Features used:

- Recorded signals: 32-channel 512Hz EEG (preprocessed to 128 Hz)
- Self-assessed emotion ratings (1-9): valence, arousal, dominance, liking (1–9)

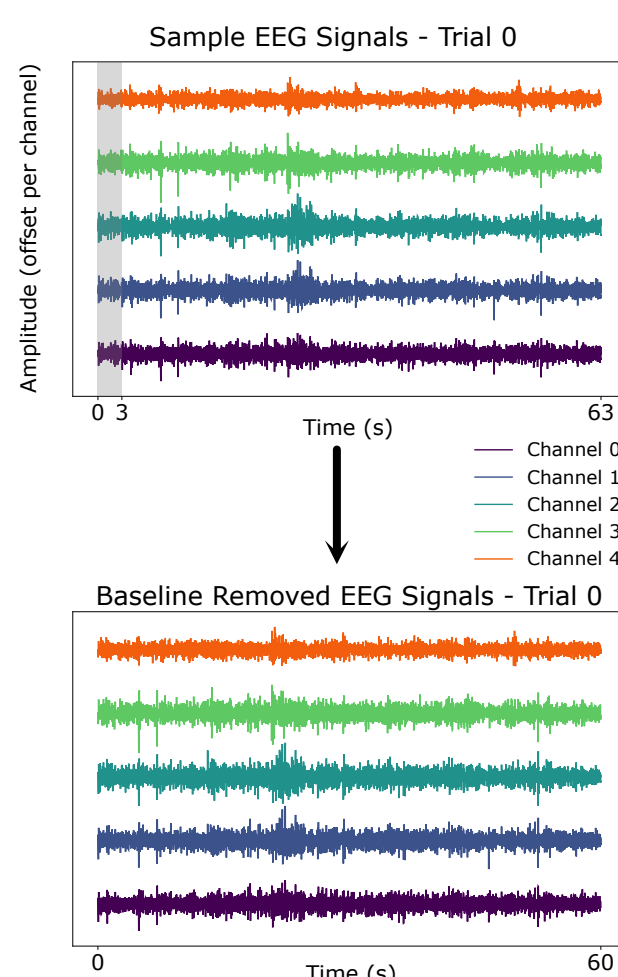
Method/Pipeline

- preprocessing
- feature extraction
- Model setup + model training
- Validation and accuracy calculation

Preprocessing and Feature extraction

1. Baseline Removal

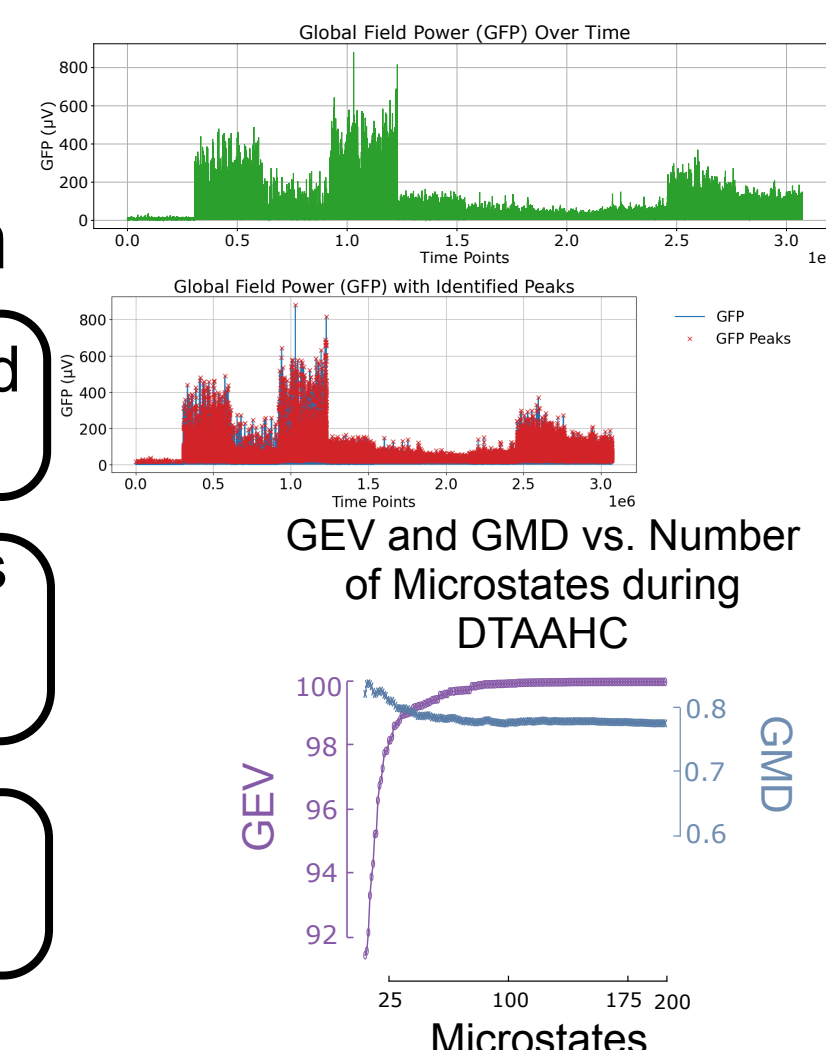
- Calculate the Mean Baseline Signal of first 3 seconds
- Segment Trial EEG Signals into 1-second, non-overlapping windows.
- Subtract mean baseline level for each 1s segment.
- concatenate segments



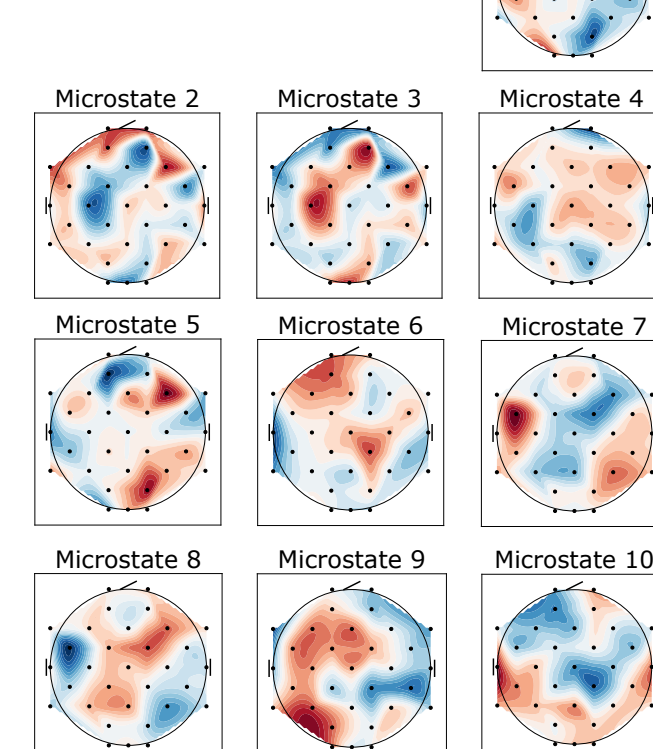
2. Segmenting into 3-second Non-overlapping Windows

3. Spatial feature extraction

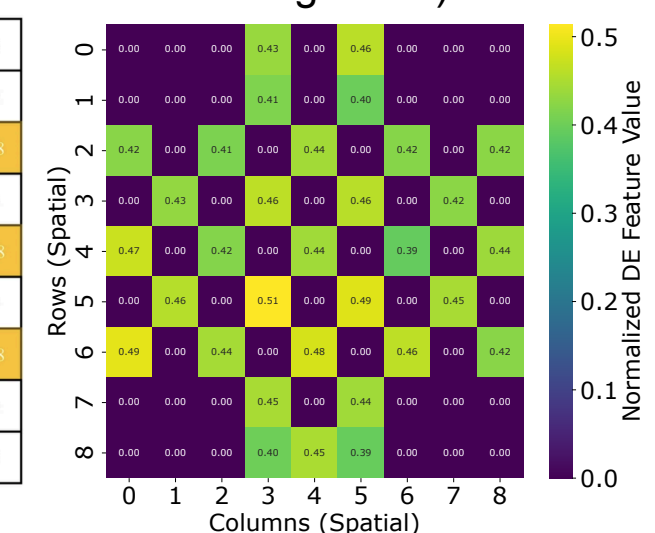
- Global Field Power (GFP), i.e. standard deviation over time
- Identify peaks and perform a K-Means pre-clustering to reduce peaks (over 100,000) to 200
- DTAAHC algorithm on K-Means centroids



Spatial topographies for 10 microstates of electric potentials across scalps

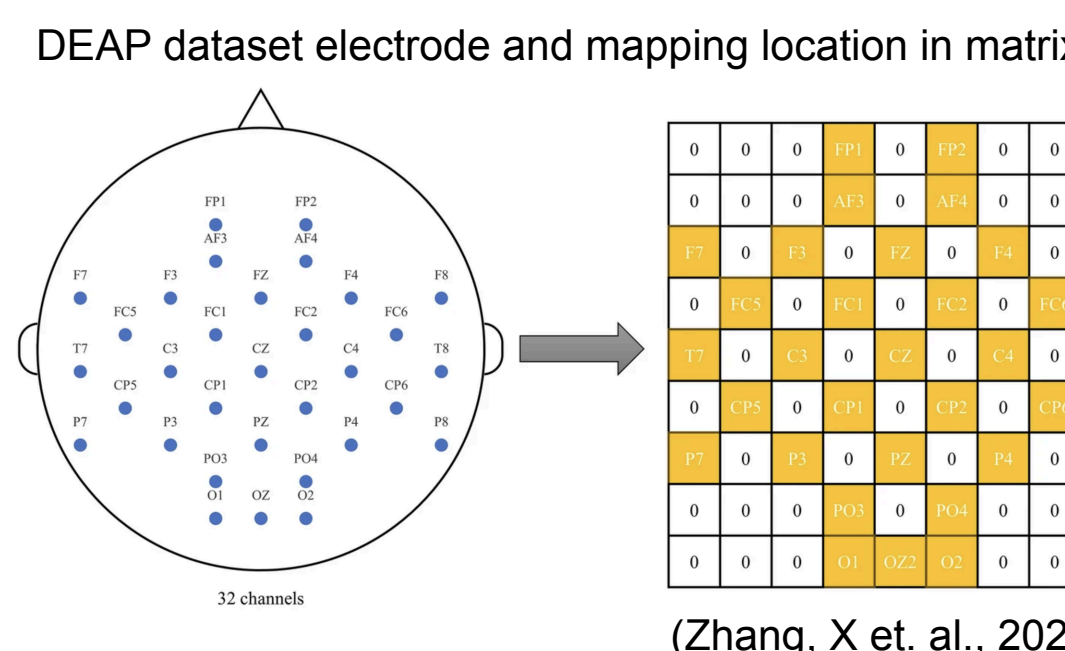


Average 9x9 Spatial Matrix (Theta Band, Original Mapped, 600 Segments)



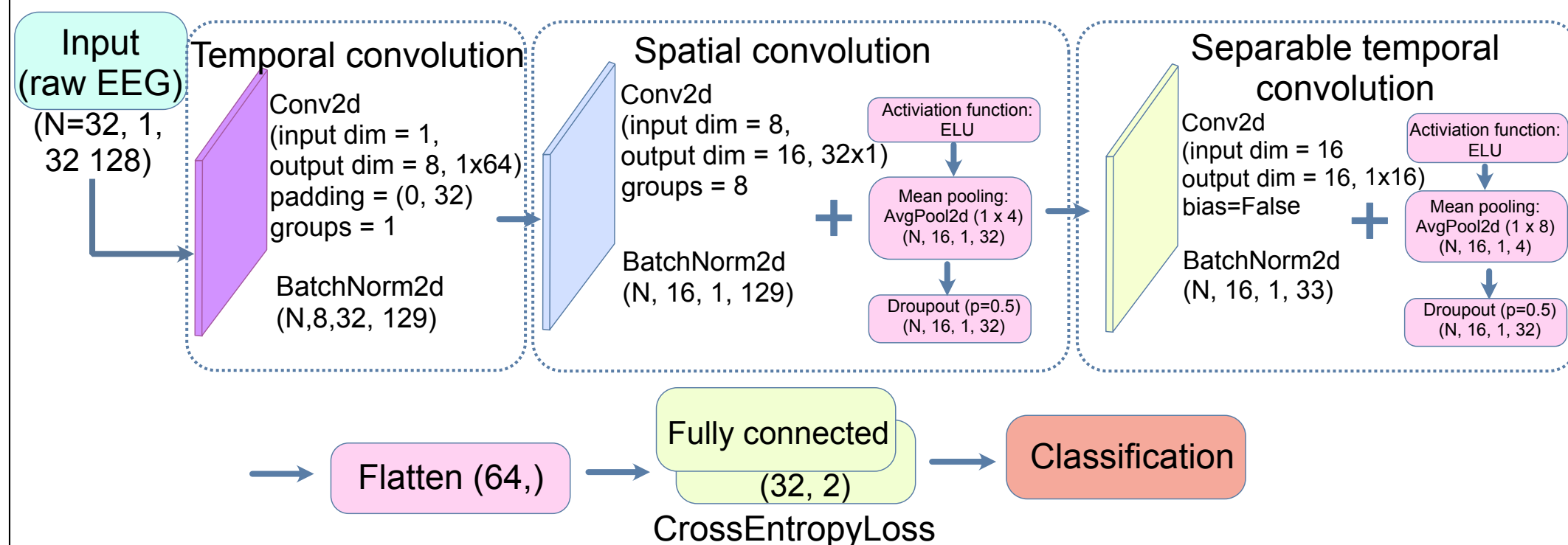
4 Temporal feature extraction

- Filter data for $\theta, \alpha, \gamma, \beta$ bands
- $DE(X) = \frac{1}{2} \log(2\pi \times e^{(\sigma^2)})$
- Normalize features using min-max scaling



Raw EEG feed into EEG net (CNN) high accuracy in Valence emotion detection

Architecture of baseline CNN



Taining loss curves for VADL

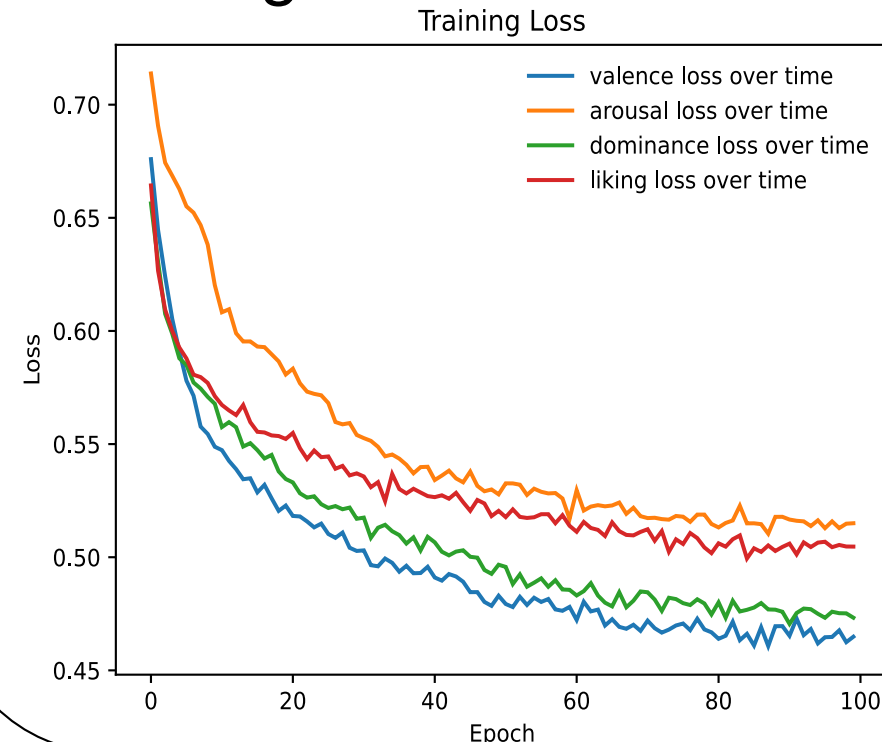


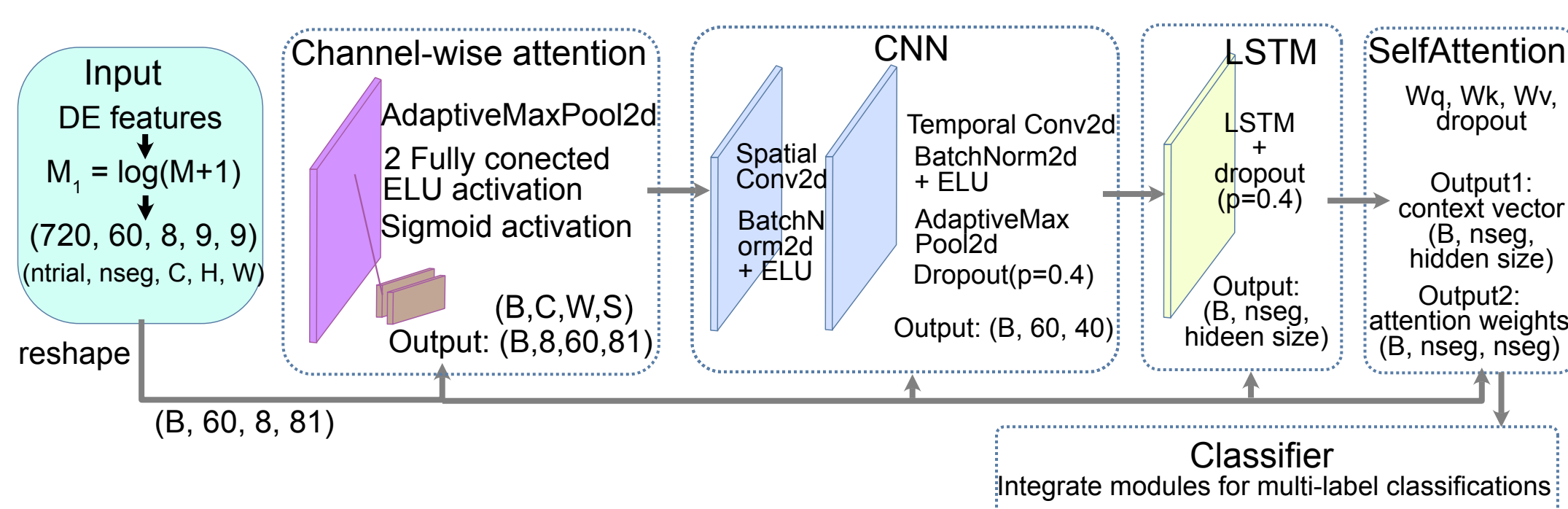
Table of calculated accuracy measurements

	Valence	Arousal	Dominance	Liking
Training accuracy	0.884	0.829	0.858	0.828
Validation accuracy	0.609	0.617	0.733	0.659
MAE	0.416	0.433	0.348	0.405
RMSE	0.517	0.514	0.421	0.483

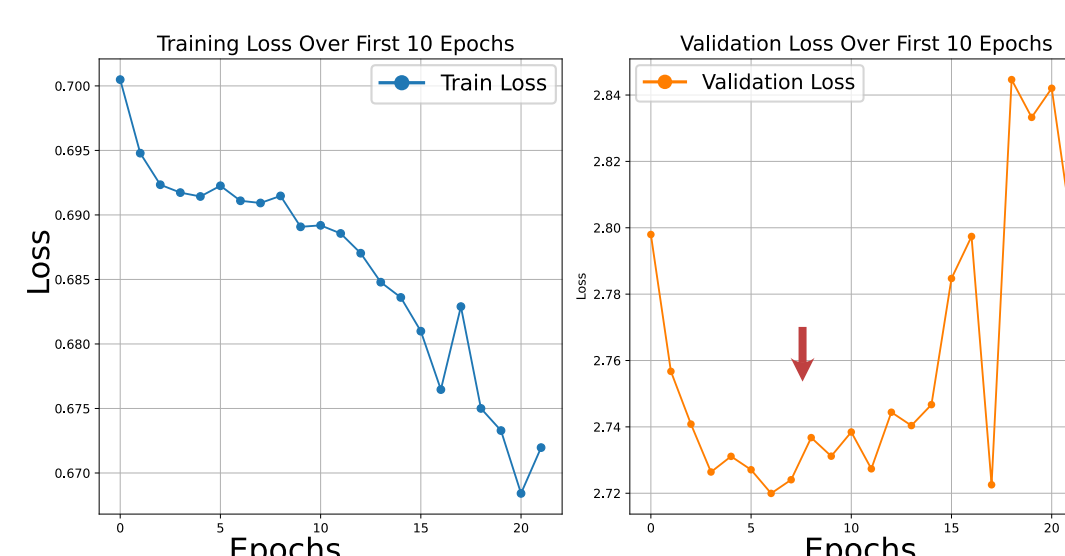
The overall accuracy is around ~80% training, ~60% validation

Attention-based convolutional recurrent neural network - ACRNN

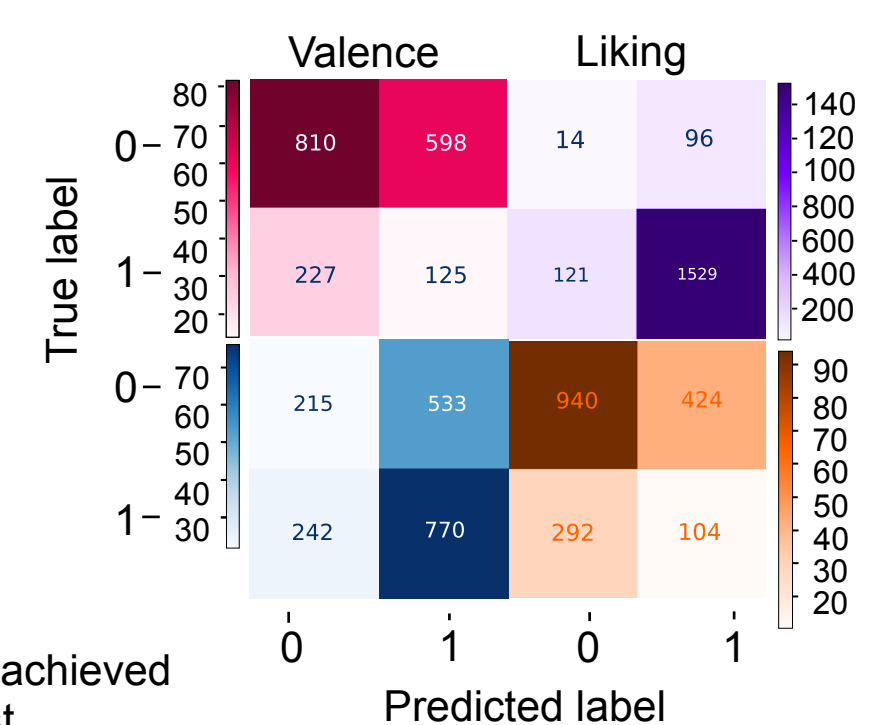
Architecture of baseline ACRNN



Training and Validation Loss curve across all 22 epochs



Confusion matrix for binary classification task across VADL

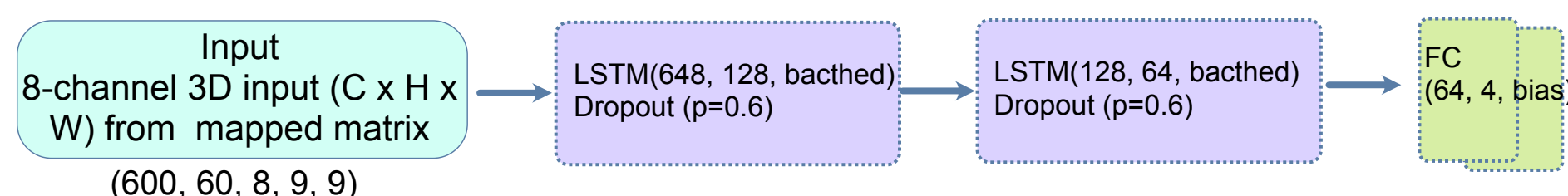


	Valence	Arousal	Dominance	Liking
Validation accuracy	0.812	0.775	0.575	0.938
MAE	0.4688	0.4068	0.4403	0.1233
RMSE	0.6847	0.6378	0.6636	0.3511

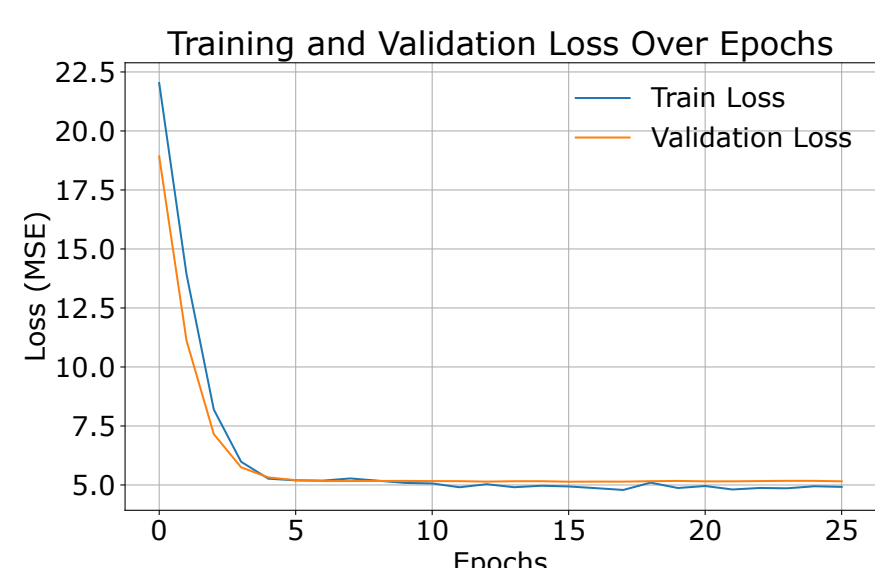
Liking achieved highest validation accuracy, MAE and RMSE

RNN classification using DE features

Architecture of proposed RNN



Validation and accuracy calculations



Valence achieve lowest RMSE across four emotions

	Valence	Arousal	Dominance	Liking
Accuracy (within 1 unit)	0.3667	0.3833	0.2250	0.2083
MAE	1.7448	1.8401	1.9212	1.9757
RMSE	2.1595	2.3502	2.2146	2.3338

Takeaways

- Baseline CNN shows that Valence classification achieved highest accuracy
- RNN taking in DE features shows highest accuracy for Valence as well.
- ACRNN achieved highest accuracy ~93% in liking classification.

Reference

- Zhang, X., Cheng, X. & Liu, H. TPRO-NET: an EEG-based emotion recognition method reflecting subtle changes in emotion. Sci Rep 14, 13491 (2024). <https://doi.org/10.1038/s41598-024-62990-4>
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